AI SMART MIRROR USING RASPBERRY PI 3B+

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ABSTRACT

In a fast-paced world where time has become a major constraint, every minute counts for busy people. With interactive computing and systems that we use in our daily lives, lifestyle and quality are changing dramatically. and products based on this interactive technology were introduced to the world. The AI Smart Mirror is a project the functionality of a traditional mirror with the intelligence of artificial intelligence. The project uses a Raspberry two-way mirror, a display, a camera, and various sensors to create a smart mirror that can display informa assistance, and improve everyday routines. The mirror can display the time, weather, news, and other personalize It can also provide assistance through voice commands, facial recognition, and gesture recognition. The project can homes, offices, and public spaces. Users can interact with the mirror by issuing commands, allowing for a mo experience.

Keywords: *AI Smart Mirror, Raspberry Pi 3B+, Two-way mirror, Face Recognition, Voice Assistant, Motion Detection.*

1. INTRODUCTION

In the present world, time is a precious commodity, particularly for busy individuals. The advent of interactive c embedded systems has revolutionized our lifestyles and dramatically improved our quality of life. These technologies rise to a wide range of devices and products that offer convenience, efficiency, and personal security. Whether at work, artificial intelligence and interactive computing have made business operations and multimedia entertainm easy for users.

One area where interactive technology can significantly impact our daily lives is in our grooming routines. People considerable time in front of mirrors, assessing their appearance, and performing various grooming tasks. This is opportunity to accomplish important activities. Enter the Smart Mirror—a remarkable innovation for busy ind embrace multitasking and desire real-time updates. The Smart Mirror is not just an ordinary mirror; it incorporates display the date, time, weather updates, daily news headlines, which makes it easier to track tasks on the go.

The Capstone project aims to develop an intelligent mirror equipped with facial recognition, a voice assistant, mot and other fundamental functions. The mirror's ability to recognize faces and detect motion improves user in personalization, making it a valuable tool for home automation, office hubs, and interactive smart environments.

2. APPROACH AND METHODOLOGY

2.1 Technology Stack

- **Raspberry Pi 3B+:** Core single-board computational platform managing sensory input-output (GPIO) and processing.
- **Python:** Primary programming language for developing automation scripts, API requests, and voice libraries.
- **OpenCV:** Computer vision framework enabling facial capture, user recognition, and motion algorithms.
- **Natural Language Processing (NLP):** Responsible for voice parsing, intent classification, and command response routing.
- **Sensor Integration:** Uses PIR sensors to trigger display awakening upon presence detection for low-overhead consumption.

2.2 Methodology and Key Features

The project follows an iterative development cycle. Steps include setting up Raspbian OS, configuring netwo building user configurations using HTML/CSS/JS interface, implementing voice engine with speech algorithms, in motion sensors, testing breathing cues, and running benchmarks. Daily updates serve news API endpoints, w in-mirror compliments, and dynamic calendar synchronizations.

3. FINAL OUTPUT & RECOGNITION RESULTS

Our fully realized Smart Mirror showcases structured columns for date, time, and custom loc notifications. When users step in path of the PIR motion sensor, the digital output wakes ins offline OpenCV facial classifier returns localized welcoming feedback (e.g. 'Good morning, with custom widgets indexing Spotify music loops, respiratory support guides, and bullet headlines.

4. CONCLUSION & FUTURE WORK

The AI Smart Mirror provides a lightweight, highly responsive, and modular computing platform smart spaces. It successfully delivers an easy-to-use, multitasking command deck, maxim efficiency without excessive overhead. Face recognition models provide accurate person i creating a personalized interface suitable for smart-homes and modern workspaces.

Future scopes can expand custom control systems, implement tactile touchscreen sensors secure smart-home dashboards (HomeAssistant), and create a unified companion mobile app ordering and personal profile preferences.

5. REFERENCES

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